

Highlights

Surrogate Learning of Scour-critical Bridge Capacities and Uncertainty Quantification Considering Climate Scenarios

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- A distribution-agnostic framework is proposed to assess scour-critical bridge capacity under climate-driven deep uncertainty.
- Surrogate models are trained using nonlinear pushover simulations with climate-scenario-based scour inputs.
- Credal-set uncertainty quantification is introduced to provide bounded predictions, ensuring objective capacity estimation.
- Results support resilient decision-making for infrastructure adaptation and asset management under uncertain climate futures.

Surrogate Learning of Scour-critical Bridge Capacities and Uncertainty Quantification Considering Climate Scenarios

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Abstract

Scour-critical bridges are among the most vulnerable transportation infrastructure systems, and their failure risk is expected to be increasingly uncertain due to climate change and climate-driven flood hazards. Traditional approaches often overlook the effects of such deep uncertainties and compute scour hazards assuming a single probabilistic distribution to analyze bridge capacity and collapse potential. This study develops a distribution-agnostic framework to predict the transverse capacities and quantify the uncertainties of scour-critical bridges. The major components include: (1) surrogate modeling of bridge capacities through learning from nonlinear pushover-based simulated data considering climate-scenario-based scour profiles, and (2) credal-set quantification of capacity uncertainties subject to climate-scenario scour profiles. The results reveal how variable uncertainties propagate through the nonlinear soil–foundation–structure bridge system when

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expressing the capacity tuple, including base shear, deck displacement, base moment, and column rotation. The resulting framework provides a novel analytics toolkit for climate-change and uncertainty-aware assessment of scour-critical bridge capacity, supporting engineers and transportation agencies in making resilient asset management decisions.

Keywords: Scour, Bridge Capacity, Surrogate Modeling, Climate Scenario, Credal Set, Soil–foundation–structure Interaction,

1. Introduction

Waterway bridge structures serve as critical nodes in transportation infrastructure, enabling vital socioeconomic connectivity. However, they are especially vulnerable to scour: the erosion of soil surrounding bridge foundations caused by flowing water. In the United States, over 600,000 bridges span the nation’s waterways, and an estimated 70% were not originally designed to resist scour. Of these, approximately 21,000 are classified as *scour-critical* (USGS, 2018). Scour has historically been the leading cause of highway bridge failure, accounting for more than half of such collapses nationwide (Simpson and Byun, 2019). Looking ahead, climate change is projected to intensify scour risk in the 21st century, as more frequent and severe riverine floods accelerate foundation erosion (Cranston and Smith, 2022). This convergence of inadequate original design, material aging, and worsening hydrologic hazards due to climate change highlights the need for predictive, risk-informed assessments of scour-critical bridges throughout their service life.

Developing reliable life-cycle performance quantification for scour-critical

18 bridges is technically challenged by numerous factors. Among these, bridge
19 scour as a hazard differs fundamentally from other natural and technological
20 hazards (e.g., earthquakes and barge impacts). One key distinction lies in
21 the uncertainties associated with scenario-based climate projections that are
22 inherent in mainstream climate modeling methodologies, whether based on
23 global climate models or downscaled regional models (Alizadeh, 2022; O'Neill
24 et al., 2020). Researchers have long referred to such uncertainties as deep
25 uncertainties (DUs), attributing them to hard-to-quantify or unknown fac-
26 tors such as long-term socioeconomic development, emission policy decisions,
27 and technological change (Kandlikar et al., 2005). The Intergovernmental
28 Panel on Climate Change (IPCC) addresses these uncertainties by employ-
29 ing multiple climate scenario frameworks (e.g., Representative Concentration
30 Pathways) that span optimistic to pessimistic emission trajectories. When
31 adopted in hazard modeling (e.g., precipitation), these climate-driven DUs
32 propagate through regional hydrological and hydraulic models (e.g., basin
33 and riverine floods), and ultimately influence local scour profiles at water-
34 way bridges.

35 Although researchers have acknowledged the complexity and implications
36 of climate scenarios and deep uncertainties (DUs) for civil infrastructure de-
37 sign and management (Helmrich and Chester, 2022), few studies have explic-
38 itly examined how these uncertainties affect bridge safety or design. Only
39 in recent years have efforts emerged to integrate climate model outputs into
40 scour risk assessments. For instance, Yang et al. linked downscaled climate
41 data to bridge failure probabilities (Yang and Frangopol, 2019), and addi-
42 tional relevant studies are reviewed below. Nevertheless, a systematic treat-

43 ment of scenario-driven uncertainty in bridge performance remains limited.
44 Several investigations have incorporated climate-scenario information into
45 scour or bridge risk modeling, yet significant gaps persist. Khelifa et al.
46 quantified the influence of intensified flood regimes on scour-failure probabil-
47 ities and economic losses; however, their framework employed deterministic
48 structural limits and did not propagate climate-projection uncertainty into
49 structural capacity metrics (Khelifa et al., 2013). Dikanski et al. explored
50 the interaction between climate-model variability and asset-data uncertainty
51 for a U.K. rail-bridge inventory, revealing that inadequate inventory data
52 can overshadow climate uncertainty by an order of magnitude. Still, their
53 analysis focused on the network level and did not address component-level
54 performance or surrogate modeling (Dikanski et al., 2018). Liu et al. de-
55 veloped a network-level, risk-based adaptation framework for bridges facing
56 climate-aggravated scour, but their approach relied on computationally ex-
57 pensive Monte Carlo simulations and was limited to discrete scenarios (Liu
58 et al., 2020). Nasr et al. generated a national-scale map of Swedish catch-
59 ments vulnerable to future pier scour but did not translate increased scour
60 depths into structural safety margins (Nasr et al., 2023). Sasidharan et al.
61 proposed a “warning-time-to-failure” index for climate-informed scour man-
62 agement while omitting probabilistic modeling of structural capacity or yield-
63 based metrics (Sasidharan et al., 2023). Collectively, these studies highlight
64 the need to model the systems-level behavior and scour risk considering cli-
65 mate scenarios; nonetheless, they do not focus on the intrinsic properties
66 (namely, capacities) of scour-critical bridges.

67 In Bridge Engineering practice, load rating has been a traditional ap-

68 proach for evaluating the vertical capacity of in-service bridges (Cai and
69 Shahawy, 2003; Scott et al., 2008; Gokce et al., 2011). This procedure typi-
70 cally involves structural analysis based on LRFD-prescribed combinations of
71 dead and live loads, hence defining the vertical capacity of bridges. For the
72 assessment of waterway bridge collapse potential, the lateral capacity rating
73 has primarily relied on nonlinear modal pushover analysis (Alipour et al.,
74 2013; Bignell et al., 2005). These methods yield nonlinear pushover curves
75 that are essential for characterizing lateral capacity under extreme transverse
76 loads, such as those induced by seismic, hydraulic, or barge impact events.
77 Capacity estimates are typically defined at the point of yielding, represented
78 by the corresponding load and displacement values. However, few studies
79 have addressed the variation in structural capacity as a function of vary-
80 ing scour depth. Moreover, no existing studies have explicitly incorporated
81 climate scenario-induced scour uncertainties. A key technical barrier in ad-
82 dressing these gaps is the high computational cost associated with modeling
83 complex nonlinear bridge systems under variable scour conditions.

84 Given the aforementioned research gaps, there is a compelling need to
85 integrate modern data-driven modeling approaches for modeling and under-
86 standing lateral capacities of scour-critical bridges with the deep uncertainties
87 inherent in climate scenarios. These machine learning (ML) methods, while
88 distinct from traditional finite element or first-principles-based approaches,
89 often remain dependent on simulation data generated through conventional
90 techniques. However, once trained on such data, the resulting ML models, i.e.
91 surrogates, can serve as computationally efficient predictive tools. Moreover,
92 by directly mapping inputs (e.g., scour depth) to outputs (e.g., structural

capacities), these models enable efficient quantification of predictive uncertainty—an essential capability when addressing scenario-based deep uncertainties central to this study.

This study proposes a novel modeling framework, which includes (1) surrogate learning of the structural capacities of scour-critical bridges under climate scenario-induced deep uncertainties and (2) quantification of climate-scenario-induced scour uncertainties without assuming a specific distribution of scour profiles. Statistically, the proposed framework is distribution-agnostic in nature. In summary, the paper presents the following key contributions:

1. *Surrogate Learning Framework for Scour Capacity.* A surrogate learning framework is developed to efficiently predict the structural capacity of scour-affected bridges using a library of nonlinear pushover analyses, enabling rapid capacity estimation without the need for repeated high-fidelity simulations.
2. *Capacity Tuple Definition.* A multi-dimensional capacity tuple is introduced to capture critical performance measures at yielding, including: (i) base shear, (ii) deck displacement, (iii) pile-cap moment, and (iv) pile-cap rotation—collectively characterizing both superstructure and substructure limit states.
3. *Climate-Induced Deep Uncertainty Quantification.* Deep uncertainties are incorporated via a credal-set approach, providing upper and lower bounds on capacity predictions across various climate scenario trajectories and yielding a robust envelope of predictive outcomes.

In the following, this paper starts with a literature review of related efforts

118 and techniques used in this work. Subsequently, it presents the proposed
119 surrogate learning and uncertainty quantification framework. The numerical
120 results given the select climate-scenario scour hazards and a representative
121 bridge model are presented. The paper further discusses the implications,
122 limitations, and directions for future research, followed by the conclusion.

123 **2. Related Efforts and Techniques**

124 *2.1. Related Efforts*

125 Numerous studies have addressed the impact of variable scour profiles
126 within fragility modeling frameworks. In some approaches, scour depth is
127 treated as a conditional variable; in others, a fully probabilistic treatment
128 is adopted, modeling scour as a hazard with an assigned distribution. Since
129 scour not only acts as a primary hazard but also alters the boundary condi-
130 tions at the foundation level, it is frequently analyzed in combination with
131 other extrinsic natural hazards, resulting in multi-hazard assessment frame-
132 works. Within this domain, several key findings have emerged:

- 133 • *Scour + Seismic.* Wang et al. (Wang et al., 2014) showed that even
134 moderate scour depths (< 2 m) can double the probability of column
135 yielding during design-level earthquakes. Guo and Chen (Guo and
136 Chen, 2015) extended this work to a life-cycle framework, while Guo et
137 al. (Guo et al., 2016) developed time-dependent fragility functions that
138 shift toward lower peak ground accelerations (PGAs) as scour severity
139 increases. Parametric and copula-based approaches (e.g., Argyroudis
140 and Mitoulis (Argyroudis and Mitoulis, 2021)) corroborate these find-
141 ings across various bridge typologies.

- 142 • *Scour + Barge Impact.* Kameshwar and Padgett (Kameshwar and Pad-
143 gett, 2018) incorporated scour-dependent lateral support loss into ves-
144 sel collision models, finding that collapse probability increases sharply
145 once scour depth exceeds roughly 1.5 pier diameters. Fan et al. further
146 integrated corrosion effects and nonlinear soil–pile interaction, reveal-
147 ing a shift in dominant failure mode from the superstructure to the
148 pile foundation (Fan et al., 2021). Guo et al. (2023) developed impact
149 fragility analysis to characterize bridge failure under barge collision and
150 local scour hazards.
- 151 • *Scour + Flood Debris / Flow.* Anisha et al. (Anisha et al., 2022)
152 jointly modeled debris impact, inundation pressure, and scour, con-
153 cluding that a 1.5 m scour depth increases flood-induced failure prob-
154 ability by approximately 80%. Karriqi et al. (Karriqi et al., 2024)
155 investigated sequential loading scenarios (earthquake followed by flood-
156 induced scour), observing moderate but systematic increases in system-
157 level fragility.

158 Despite variations in modeling strategies, these studies share a common
159 approach to probabilistically estimating structural collapse. For a generic
160 structure, this begins by defining a displacement-based capacity measure:
161 typically, the yielding displacement denoted Δ_y , obtained via nonlinear pushover
162 analysis. A corresponding response-demand quantity, Δ , is then computed
163 by simulating structural response under hazardous forces (e.g., from seismic
164 or barge-impact loadings) characterized by an intensity measure H . The col-
165 lapse event is then defined as the normalized demand $u = \Delta/\Delta_y$ exceeding a

threshold value u_t , namely $u > u_t$. Through stochastic simulation that incorporates load and material uncertainties, researchers estimate the conditional probability of collapse, $P(u > u_t | H)$, namely collapse fragility. For conventional structures with fixed boundary conditions, Δ_y is treated as a constant property of the structure. However, in scour-critical bridges, Δ_y varies with scour depth, as scour alters the boundary conditions at the foundation level. Furthermore, it introduces ambiguity in selecting an appropriate threshold constant, u_t .

To illustrate this concept, Figure 1 presents three nonlinear pushover profiles corresponding to different scour conditions. The solid curve, with yielding capacity (Δ_{y0}, V_{y0}) , represents the non-scour case. The dashed curve with (Δ_{y1}, V_{y1}) corresponds to a moderate scour scenario, and the dot-dashed curve with (Δ_{y2}, V_{y2}) reflects a severe scour condition. If a constant capacity, such as Δ_{y0} , is used to evaluate collapse potential across all scenarios, the resulting index $u = \Delta/\Delta_{y0}$ would misleadingly suggest equal vulnerability to collapse for the underlying bridge with the varying scour conditions. This highlights a fundamental limitation in using fixed capacity metrics under variable scour conditions. Moreover, as shown in Figure 1, both displacement and strength capacities (Δ_y and V_y) are reduced with increasing scour severity. Consequently, collapse limit states for scour-critical bridges must account for reductions in both strength and displacement capacities to reflect realistically the system capacity under scour-induced degradation.

2.2. Climate-scenario Uncertainty and Spatial Analogue

Decision-making in hydraulic and infrastructure engineering under climate change is increasingly challenged by *deep uncertainties* associated with

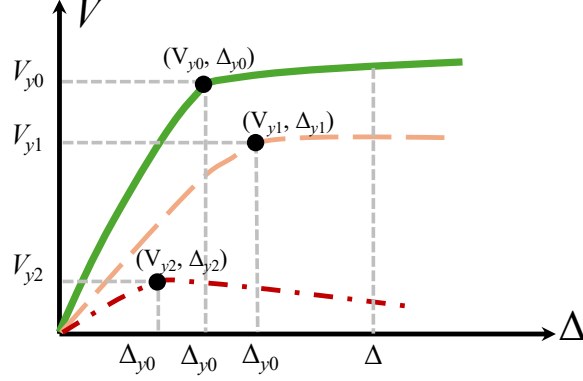


Figure 1: Illustration of representative nonlinear pushover curves for scour-critical bridge structures under varying scour profiles.

191 various climate scenarios (IPCC, 2022). This condition is particularly acute
 192 in the context of projecting localized flood hazards, where uncertainty propa-
 193 gates across multiple modeling layers, from global climate models (GCMs) to
 194 regional downscaling, and more localized hydrological simulations (Lafferty
 195 and Sriver, 2023).

196 Under such uncertainty, conventional risk-based planning methods that
 197 rely on probabilistic forecasts become insufficient. As emphasized by Lem-
 198 pert et al. (Lempert et al., 2024), deep uncertainty necessitates alternative
 199 strategies that explore a broad range of plausible futures rather than opti-
 200 mizing for a single forecast. In this context, *Robust Decision Making (RDM)*
 201 and *Dynamic Adaptive Policy Pathways (DAPP)* have emerged as leading
 202 frameworks.

203 One practical implementation of these strategies in hydrological risk as-
 204 sessment is the use of **spatial analogue**: the substitution of geographically
 205 distinct but hydrologically relevant locations to represent potential future

206 conditions at a target site. This space-for-time reasoning is rooted in the
 207 ergodic hypothesis, which asserts that spatial ensembles governed by simi-
 208 lar physical processes can approximate temporal evolution (Sivapalan et al.,
 209 2011). In hydrology, this principle supports the idea that rivers in disparate
 210 climate regimes today may serve as empirical analogues for the future be-
 211 havior of a local watershed under climate change.

212 This analogue-based scenario selection offers key advantages: it anchors
 213 scenario construction in observed extremes, supports empirical stress-testing
 214 aligned with IPCC guidance (IPCC, 2022), and enables site-specific yet trans-
 215 ferable hazard modeling. In this study, selected watersheds serve as proxies
 216 for future conditions, forming the empirical basis for generating synthetic
 217 but physically plausible flood and scour scenarios. Moreover, by bypass-
 218 ing reliance on predefined probabilistic distributions, the spatial-analogue
 219 approach directly supports a distribution-agnostic framework for modeling
 220 scour-critical bridge capacities under deep climate uncertainty.

221 *2.3. Surrogate Modeling Methods*

222 Data-driven surrogate models have emerged as efficient alternatives for
 223 simulating nonlinear structural responses under uncertainty. Among them,
 224 Support Vector Regression (SVR) and Gradient Boosting Regression (GBR)
 225 are widely used due to their complementary strengths in learning complex re-
 226 lationships from limited or structured data. SVR aims to find a function $f(\mathbf{x})$
 227 that minimizes a regularized ϵ -insensitive loss, promoting both model flatness
 228 and tolerance to small errors. This makes SVR particularly effective when
 229 training datasets are small or the underlying structural behavior is smooth.
 230 GBR, on the other hand, constructs an additive ensemble of shallow deci-

231 sion trees, where each successive tree fits the residuals of the previous one.
 232 This sequential learning strategy enables GBR to capture sharp nonlinear
 233 transitions and feature interactions effectively. Comparatively, toward gen-
 234 eralization, SVR leverages kernel-based representations to generalize well in
 235 low-data regimes, while GBR achieves high predictive accuracy by exploiting
 236 structural hierarchies in unseen data. For formal mathematical formulations
 237 of SVR and GBR, readers are referred to (Smola and Schölkopf, 2004; Fried-
 238 man, 2001).

239 Many researchers adopted SVR and GBR as surrogate modeling tech-
 240 niques modeling structural behavior in Civil Structural and Infrastructure
 241 Engineering [e.g., (Xu et al., 2020; Wang et al., 2022)]. Specifically, recent
 242 benchmarking studies, including Zhao et al. (Zhao et al., 2023), demonstrate
 243 that SVR and GBR consistently outperform linear models and often rival
 244 deep learning approaches, while requiring significantly less computational
 245 effort. These properties are especially advantageous in uncertainty quantifi-
 246 cation tasks, where repeated evaluation of surrogate models is necessary. In
 247 this study, SVR and GBR were selected as surrogate modeling techniques
 248 based on their proven performance in structural engineering applications.

249 *2.4. Uncertainty Quantification Methods*

250 Modern machine learning and statistics provide several methods to esti-
 251 mate prediction uncertainty; representative ones include:

- 252 • *Bayesian Predictive Intervals:* Bayesian approaches (e.g., Gaussian
 253 processes) yield posterior distributions and credible intervals by treat-
 254 ing model parameters as random variables (Rasmussen and Williams,
 255 2006).

- 256 • *Conformal Prediction*: Conformal methods produce intervals with fi-
 257 nite sample coverage guarantees, assuming future data follow the same
 258 distribution as training Angelopoulos and Bates (2021)
- 259 • *Credal Sets*: Rather than a single distribution, a credal set represents
 260 a set of plausible probability measures for the output, capturing both
 261 aleatory and epistemic uncertainty (Walley, 1991; Höffman et al., 2022).

262 While Bayesian and conformal intervals perform well in many practical
 263 situations, they often underestimate uncertainty when facing distributional
 264 shifts or deep epistemic uncertainty. In this work, we adopt credal set meth-
 265 ods to characterize uncertainties associated with bridge capacity prediction.
 266 This nonparametric distribution-agnostic approach yields a robust envelope
 267 that widens in regions of high epistemic uncertainty, as model disagreement
 268 increases (Breiman, 1996).

269 3. Methodology

270 3.1. Overview

271 Figure 2 outlines the overall framework, which integrates three major
 272 components:

- 273 1. Scenario Selection, Scour Hazard, and FE Model Generation. This is a
 274 preparation stage that focuses on establishing the foundational inputs
 275 for simulation, including specific hydraulic flow scenarios and a suite
 276 of scour conditions generated using probabilistic sampling. The finite-
 277 element model of the bridge of interest is systematically modified for
 278 each sampled scour condition.

- 279 2. Nonlinear Pushover Simulation and Capacity Extraction. This com-
 280 ponent executes nonlinear analyses for each scour case and extracts
 281 yield-level capacity responses. The outputs form a dataset for surro-
 282 gate model training.
- 283 3. Surrogate Modeling and Credal-Set Uncertainty Quantification (UQ).
 284 The final stage replaces computationally expensive simulations with
 285 lightweight machine learning models, enabling rapid capacity prediction
 286 and uncertainty quantification.

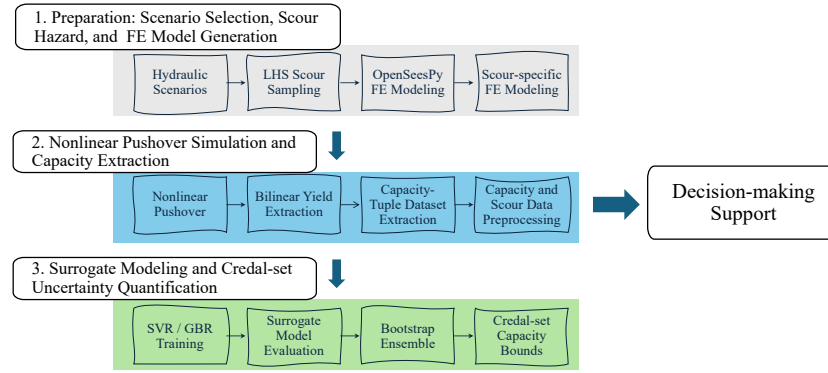


Figure 2: Proposed surrogate modeling and uncertainty quantification framework and methodological components.

287 Critical technical details and intermediate results from the climate-scenario-
 288 based scour analysis, FE modeling, and capacity value extraction are pre-
 289 sented in the following subsections. Technical methods for surrogate mod-
 290 eling and credal-set uncertainty quantification are described subsequently,
 291 followed by major result interpretations. The workflow is implemented using
 292 Python, including OpenSeesPy (Zhu, 2025) for FE simulation and Scikit-
 293 learn (scikit-learn Developers, 2025) for machine learning and uncertainty

294 quantification.

295 3.2. Scenario Selection and Scour Hazard Calculation

296 The specific design case involves a bridge spanning an imagined Midwest-
297 ern river channel. Given the difficulty of downscaling peak flow velocities
298 from global climate projections, the spatial-analogue concept is adopted in
299 this work. Each scenario corresponds to a flooding event of increasing sever-
300 ity and is tied to a potential climate future near the year 2050, as structured
301 by the SSP scenarios in IPCC AR6 (IPCC, 2022).

302 We identify two real-world rivers and one assumed worst-case as analogs
303 to capture the range of plausible future extremes (Table 1):

- 304 • *Scenario 1 – Moderate ($U = 2.9$ m/s)*: Based on average flood velocities
305 reported for the *Missouri River*, this case represents a baseline future
306 in which climate change causes moderate increases in flow velocities
307 (MDOT, 2019).
- 308 • *Scenario 2 – High ($U = 6.5$ m/s)*: Represents peak flood conditions in a
309 more extreme but still physically observed context, based on field mea-
310 surements from the *Colorado River* during a major flood event (Dawson
311 and Brunner, 2009). This scenario reflects the increased intensity and
312 flashiness expected in some AR6 pathways (e.g., SSP5-8.5).
- 313 • *Scenario 3 – Extreme ($U = 10$ m/s)*: A worst-case design flood veloc-
314 ity estimated for major Midwestern river systems under 500-year re-
315 currence conditions. This provides an upper envelope for stress-testing
316 structural safety and resilience.

317 To characterize long-term pier scour under variable hydraulic conditions,
 318 this study adopts the erosion framework from Briaud et al. (Briaud et al.,
 319 1999), widely used for cohesive soils and implemented in HEC-18. The
 320 adopted erosion rates $\dot{z}$, classified based on Briaud’s framework, are sum-
 321 marized in Table 1, corresponding to the selected scenarios and expected
 322 hydraulic conditions. It is noted that herein the notation z is adopted for
 323 scour and erosion following Briaud et al. (1999), which does not refer to the
 324 z coordinate. Later, for surrogating modeling, the symbol S_z is defined as a
 325 scour depth variable.

Table 1: Scenario-Based Hydraulic Inputs and Scour Predictions (LHS, $N = 1000$)

<i>Scenario</i>	U (m/s)	<i>Analog River (Flow Description)</i>	$\dot{z}$ (mm/h)	μ_{scour} (m)	σ (m)	95 th -Perc (m)
1	2.9	Missouri River (avg. flood)	100	3.56	1.70	6.80
2	6.5	Colorado River (2009 peak)	500	9.13	4.26	17.16
3	10.0	Midwestern extreme (500-yr design flood)	1000	14.10	6.74	26.30

326 First, the maximum hydraulic shear stress τ_{max} is defined as a function of
 327 upstream velocity U , water density ρ_w , and Reynolds number Re , computed
 328 as:

$$Re = \frac{U d_p}{\gamma} \quad (1)$$

329 where d_p is the pier diameter and γ is the kinematic viscosity. The shear
 330 stress τ_{max} is then given by the semi-empirical relation:

$$\tau_{\text{max}} = 0.094 \cdot \rho_w \cdot U^2 \left[\frac{1}{\log_{10}(Re)} - 1 \right] \quad (2)$$

331 The maximum scour depth under clear-water conditions is expressed as:

$$z_{\text{max}} = 0.18 \cdot Re^{0.635} \quad (3)$$

332 This expression is derived from regression analysis of controlled tests con-
 333 ducted in cohesive soils (Gudavalli, 1998).

334 The equilibrium scour time t_{eq} is modeled as a duration-based erosion
 335 process that scales with time t , flow velocity U , and initial scour rate $\dot{z}$:

$$t_{\text{eq}} = 73 \cdot t^{0.126} \cdot U^{1.706} \cdot \dot{z}^{-0.2} \quad (4)$$

336 From this, the projected scour depth after 50 years, denoted z_{50} , is esti-
 337 mated by:

$$z_{50} = \frac{t_{\text{eq}}}{1 + t_{\text{eq}}} z_{\text{max}} \quad (5)$$

338 To capture uncertainty in long-term scour development due to stochastic
 339 variability in material properties and hydraulic conditions, a lognormal model
 340 is used for the final scour depth. Given a standard normal variate $\epsilon \sim \mathcal{N}(0, 1)$,
 341 the final scour depth is computed as:

$$z_{50}^{\text{final}} = \frac{e^{-0.085z_{50}} \cdot e^{0.407\epsilon}}{1000} \quad (6)$$

342 where the denominator converts millimeters to meters. This probabilistic
 343 formulation aligns with previous risk-informed studies on scour and bridge
 344 resilience (Alipour et al., 2013; Briaud and Oh, 2010).

345 To approximate the distribution of introduced random variables, the
 346 Latin Hypercube Sampling (LHS) method is adopted. The scour-depth sam-
 347 ples under the three different river flow conditions (Table 1) are simulated
 348 separately. For each scenario, the script generates 1,000 samples by convert-
 349 ing flow velocity into estimated scour depth. It accounts for uncertainty in
 350 both hydraulic forces and soil erosion characteristics. Figure 3a and 3b il-
 351 lustrate the resulting distributions, including histogram plots, lognormal fits

denoted by $f_{S_z}(z)$, and the fitted exceedance probability curves representing the hazard function: $P_{S_z}(Z > z) = 1 - \int_{-\infty}^z f_{S_z}(z') dz'$.

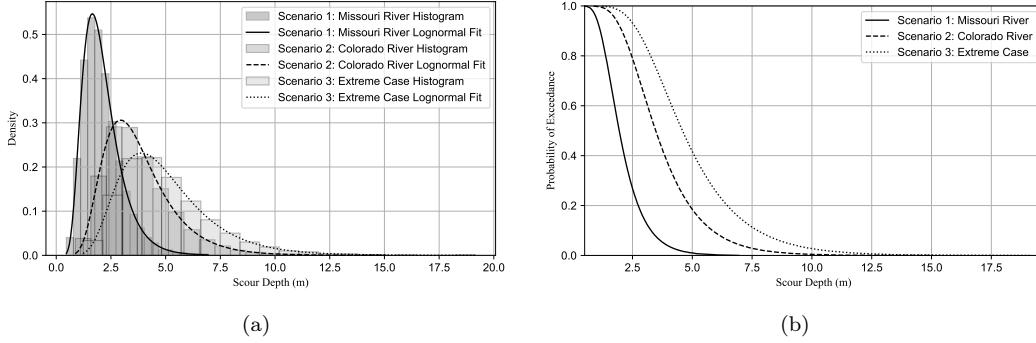


Figure 3: (a) Histogram distributions and lognormal fits for simulated climate scenario-based scour depth data. (b) Fitted hazard curves for cumulative distribution of simulated climate scenario-based scour depth data.

3.3. Finite-Element Modeling

As reviewed previously, many scour-critical bridge models have been developed in the literature. The framework proposed in this work is adaptable to any waterway bridge model, provided that climate-sensitive scour hazards are of interest. The bridge model adopted here is based on the one developed in Atak et al. (2014), originally implemented using the OpenSees framework with the TCL language. In this study, the model is re-implemented in Python using the OpenSeesPy library. Figure 4a illustrates the transverse elevation of the bridge structure, and Figure 4b provides details of the primary elements developed in the OpenSeesPy environment.

The modeled bridge represents a typical Midwestern U.S. bridge: a four-span, non-skewed prestressed concrete structure, with each span measuring 35 meters and a total deck width of 12 meters. The superstructure consists

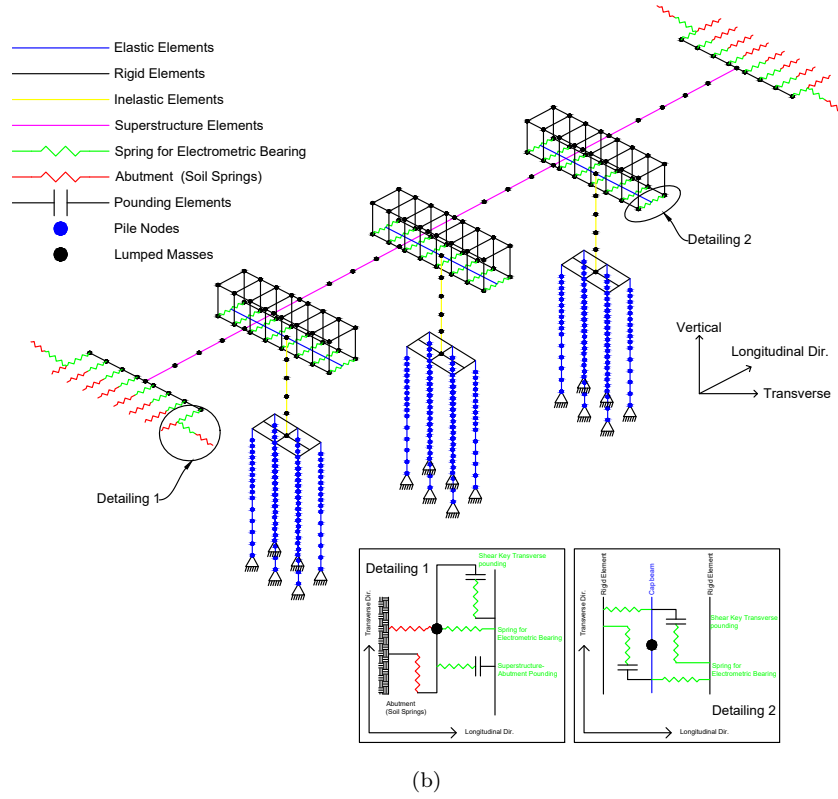
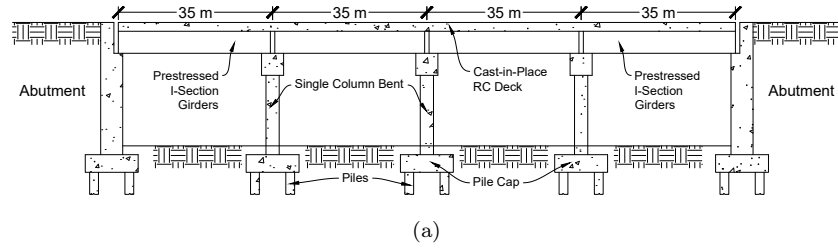


Figure 4: (a) Elevation view of the bridge structure. (b) The finite-element model with key nonlinear elements.

367 of I-girders supporting a cast-in-place concrete deck, resting on elastomeric
368 bearings over reinforced concrete piers and pile group foundations embedded
369 in layered soil.

370 The piers are modeled using nonlinear beam-column elements with fiber
371 sections to accurately capture flexural behavior, confinement, and post-yield
372 response. Each pier column is divided into five integration points, provid-
373 ing a detailed curvature distribution. The `Concrete01` material is used for
374 both unconfined and confined concrete zones, while `Steel02` is adopted for
375 reinforcing steel, with cyclic hardening parameters $R_0 = 15.0$, $c_{R1} = 0.9$,
376 and $c_{R2} = 0.2$. Shear and torsional effects are captured using Aggregator
377 sections.

378 Soil-structure interaction is modeled through nonlinear `zeroLength` springs
379 at the pile-soil interface, simulating p - y (lateral), t - z (vertical), and q - z (tip)
380 behaviors following FHWA HEC-18 recommendations. These springs use the
381 `PySimple1` material definition and are depth-dependent. Springs above the
382 scour depth are deactivated to simulate soil loss, thereby reducing lateral and
383 axial stiffness.

384 To reflect scour-induced soil loss, springs are progressively removed as
385 scour depth increases. While material property uncertainties are included,
386 they are intentionally minimized so that scour remains the dominant source
387 of variability. Specifically, concrete compressive strength f'_c is modeled as a
388 normal distribution with a mean of 27.0 MPa and a standard deviation of
389 1.0 MPa. Steel yield strength f_y is modeled using a lognormal distribution
390 with a mean of 420.0 MPa and a standard deviation of 4.2 MPa.

391 Latin Hypercube Sampling (LHS) is employed with 1,000 samples per sce-

392 nario (three scenarios in total), resulting in 3,000 simulations. This sampling
 393 approach robustly approximates the probabilistic scour hazard while keeping
 394 material variability constrained (Hwang et al., 2001). Figures 5a and 5b de-
 395 pict the resulting histograms based on the sampled values for concrete and
 396 reinforcing steel strength, respectively.

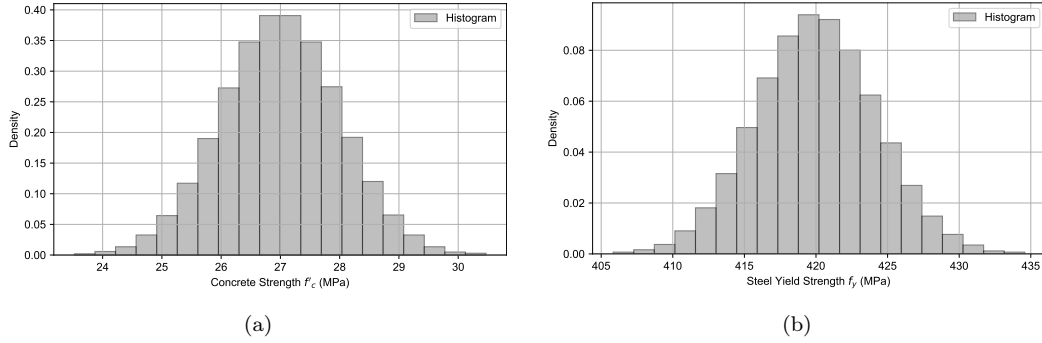


Figure 5: (a) Histogram of concrete compressive strength f'_c (MPa). (b) Histogram of
 steel yield strength f_y (MPa).

397 3.4. FE Simulation and Capacity Extraction

398 Pushover analyses were conducted for each simulation, tracking the force-
 399 displacement response at the top of the piers. A total of 3,000 nonlinear
 400 pushover simulations were performed, corresponding to Latin Hypercube
 401 Sampling (LHS)-generated scour cases across three hydraulic scenarios. For
 402 each scour case, the OpenSeesPy model was regenerated to reflect the corre-
 403 sponding scour depth by deactivating `zeroLength` soil springs (representing
 404 p - y , t - z , and q - z behaviors) located above the scour level.

405 After applying and equilibrating gravity loads, a displacement-controlled
 406 pushover analysis was performed in the longitudinal direction ($\text{DOF} = 2$)
 407 at the deck level, targeting a maximum lateral drift of 5% of the column

height. A fallback strategy was implemented to ensure convergence, adaptively switching among Newton–EnergyIncr, Broyden, NewtonLineSearch, and Krylov-based solvers. Each simulation recorded top-node displacements, column rotations, base shears, and moments. The key structural responses at yielding were extracted and organized into a four-dimensional capacity tuple:

- *Yield Displacement* (Δ_y): The horizontal displacement at the bilinear yield point, obtained from curve fitting.
- *Yield Base Shear* (V_y): The lateral force corresponding to Δ_y .
- *Yield Moment* (M_x): The bending moment recorded at the base of the critical columns at yield.
- *Yield Rotation* (θ_x): The rotational deformation at yielding, computed from column displacement profiles.

This set of yield capacities (Δ_y , V_y , M_x , θ_x) captures essential limit-state behaviors and supports cross-scenario performance comparisons. Each pushover response (f vs. d) was smoothed and fitted using a segmented bilinear function:

$$\hat{f}(d) = \begin{cases} k_1 \cdot d, & d < d_y \\ k_2 \cdot d + (k_1 - k_2) \cdot d_y, & d \geq d_y \end{cases} \quad (7)$$

A profile-fitting algorithm (see Listing 1) minimized the squared residuals between the simulation output f_i and the predicted values $\hat{f}(d_i)$:

$$J(k_1, k_2, d_y) = \sum_{i=1}^N \left[f_i - \hat{f}(d_i; k_1, k_2, d_y) \right]^2 \quad (8)$$

427 The optimal yield displacement d_y was selected by sweeping over a grid
 428 of candidate breakpoints. For each d_y , the best-fit values of k_1 and k_2 were
 429 computed using closed-form expressions:

$$k_1 = \max \left\{ 0, \frac{\sum_{i \in I_1} d_i f_i}{\sum_{i \in I_1} d_i^2} \right\}, \quad k_2 = \frac{\sum_{i \in I_2} (d_i - d_y)(f_i - k_1 d_y)}{\sum_{i \in I_2} (d_i - d_y)^2} \quad (9)$$

430 Figure 6 shows an example of bilinear fitting for one simulated scour
 431 depth. The yielding base shear and the corresponding deck-level displace-
 432 ment are extracted from this fit.

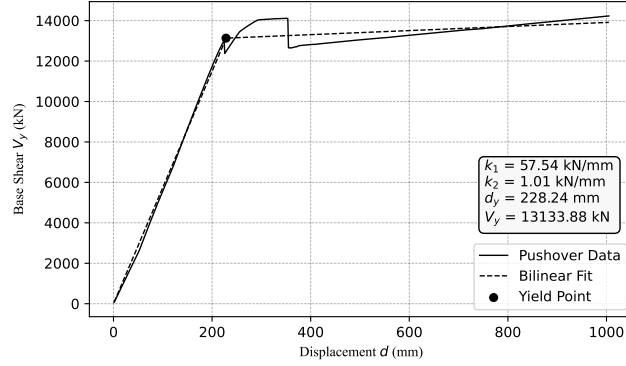


Figure 6: Illustration of capacity-tuple extraction using bilinear pushover curve fitting to extract Δ_y and V_y at a scour depth of 7.347 m.

433 Figure 7 presents histogram plots for the extracted capacity values of all
 434 3,000 simulations, combining the results from all climate scenarios: (a) V_y ,
 435 (b) M_x , (c) Δ_y , and (d) θ_x . A key observation is that the resulting distribu-
 436 tions deviate significantly from common parametric forms—exhibiting both
 437 skewness and multimodality. This indicates that fitting these distributions
 438 to standard probability models (e.g., normal or lognormal) is not meaning-
 439 ful, thereby reinforcing the need for nonparametric uncertainty quantification
 440 methods.

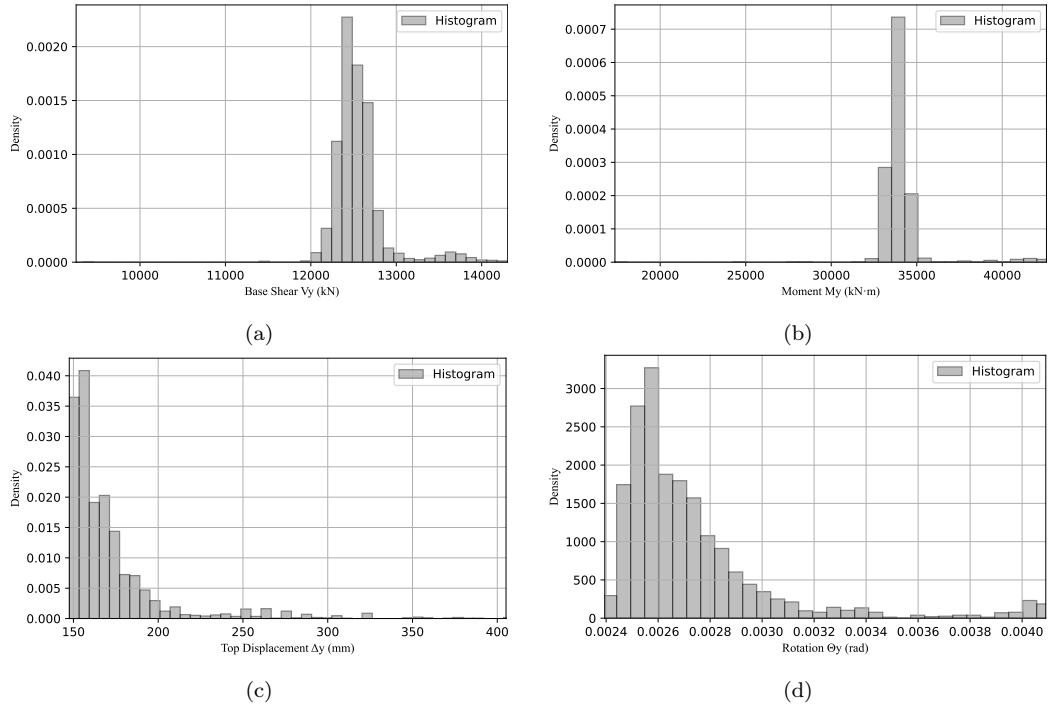


Figure 7: Resulting histogram plots for the bridge capacity tuple: (a) V_y , (b) M_x , (c) Δ_y , and (d) θ_x .

441 3.5. Surrogate Modeling

442 This study develops data-driven surrogate models to predict structural
 443 capacity measures directly from scour depth input. The goal is to replace
 444 computationally intensive finite-element simulations with lightweight predic-
 445 tive models mapping scour depth (S_z) to the four components of the capacity
 446 tuple:

$$S_z \xrightarrow{\text{Predict}} \Delta_y, V_y, M_x, \theta_x \quad (10)$$

447 where Δ_y is the lateral displacement at yielding, V_y is the base shear, M_x is
 448 the flexural moment, and θ_x is the rotational deformation at yield. These sur-
 449rogate models are trained using output from nonlinear pushover simulations,
 450 where each data point links a specific scour condition to a corresponding
 451 capacity tuple.

452 To enrich the input space and improve the model’s ability to approximate
 453 nonlinear patterns, we engineer features based on S_z : S_z^2 , S_z^3 , $\log(S_z)$, $1/S_z$,
 454 and $\sqrt{S_z}$. These transformations help the model capture both polynomial
 455 and asymptotic trends in capacity loss under increasing scour conditions.

456 Before model training, data were cleaned to remove incomplete or unsta-
 457 ble simulations. An outlier filtering step was applied within scour-depth bins
 458 using the interquartile range (IQR) method: points lying outside $[Q1 - 1.5 \times$
 459 $IQR, Q3 + 1.5 \times IQR]$ were discarded (Tukey, 1977). This localized filtering
 460 prevented heavily scoured samples from distorting the statistical characteri-
 461 zation of lightly scoured cases, and vice versa.

462 3.5.1. *Support Vector Regression (SVR).*

463 Support Vector Regression (SVR) is a kernel-based learning algorithm
464 that approximates a nonlinear function mapping input features to target
465 responses. It projects the input vector $\mathbf{x}$ into a high-dimensional space using
466 a kernel function (here, the Radial Basis Function or RBF kernel) and finds
467 a function that fits the training data while ignoring small errors within a
468 specified tolerance band. SVR is especially effective when the training data
469 are limited and smooth nonlinear relationships exist.

470 Key hyperparameters in SVR include: the penalty parameter C , which
471 controls the trade-off between model flatness and tolerance to training er-
472 rors; the ϵ -insensitive zone, which defines the range within which errors are
473 ignored; and the kernel coefficient γ , which governs the influence of individ-
474 ual training samples in the feature space. For mathematical formulations
475 and detailed derivation, see (Smola and Schölkopf, 2004).

Table 2: Combined SVR and GBR hyperparameter settings used in surrogate modeling.

SVR		GBR	
<i>Parameter</i>	<i>Value</i>	<i>Parameter</i>	<i>Value</i>
Kernel	RBF	Number of Estimators	100
ϵ (insensitive zone)	0.01	Learning Rate (ν)	0.05
C (penalty term)	10.0	C_{ub} Tree Depth	3
γ (kernel coefficient)	auto	C_{lb} Samples per Leaf	5
Number of Bootstrap Models	30	Subsample Ratio	0.8
—	—	Number of Bootstrap Models	30

476 *3.5.2. Gradient Boosting Regression (GBR).*

477 Gradient Boosting Regression (GBR) is an ensemble-based method that
478 builds a predictive model by sequentially adding weak learners, typically
479 shallow regression trees, each of which attempts to correct the residuals of
480 the combined model thus far. This additive process enables the ensemble to
481 capture complex nonlinear relationships in the data.

482 Important GBR hyperparameters include: the number of estimators (i.e.,
483 trees), the learning rate ν which moderates how much each tree contributes to
484 the final prediction, the maximum tree depth which controls model complex-
485 ity, the minimum number of samples per leaf which helps prevent overfitting,
486 and the subsample ratio which governs the fraction of samples used per tree.
487 For mathematical details on gradient boosting, see (Friedman, 2001).

488 *3.5.3. Hyperparameters and Performance Metrics*

489 Hyperparameter selection for both SVR and GBR was conducted via grid
490 search with 5-fold cross-validation using bootstrapped datasets. Specifically,
491 the optimisation objective was to minimise the mean squared error (MSE)
492 across validation folds. For SVR, the penalty parameter C and kernel pa-
493 rameter γ were varied within logarithmic ranges $[10^{-2}, 10^3]$, while for GBR,
494 the learning rate and maximum tree depth were tuned within $[0.01, 0.2]$ and
495 $[2, 6]$, respectively. Other parameters followed the default settings of the
496 `scikit-learn` implementation. The final hyperparameter settings are pro-
497 vided in Table 2.

498 To assess the predictive performance of the trained surrogate models
499 (SVR and GBR), we employ two widely used evaluation metrics on a held-out
500 test dataset and adopt two classic measures: (1) Coefficient of Determination

501 (R^2) , which measures the proportion of variance in the dependent variable
502 that is predictable from the input variables, and (2) Root Mean Square Error
503 (RMSE), which is a measure of the average magnitude of prediction errors.
504 The two metrics together provide a balanced view of both model accuracy
505 and variance-explaining capability across the different components of the
506 structural capacity tuple.

507 3.6. Credal-set Uncertainty Quantification

508 A credal set can be defined in its simplest form as an interval prediction: $f(\mathbf{x}) \in$
509 $[\underline{f}(\mathbf{x}), \bar{f}(\mathbf{x})]$, or more generally as a convex set of distributions $K(\mathbf{x})$ over
510 the output $f(\mathbf{x})$ given input $\mathbf{x}$, where $K(\mathbf{x})$ denotes the set of all plausible
511 probability measures consistent with available data and modeling assump-
512 tions (Walley, 1991). The interval bounds $\underline{f}(\mathbf{x})$ and $\bar{f}(\mathbf{x})$ represent the lower
513 and upper envelopes of predicted outcomes, capturing epistemic uncertainty
514 from data scarcity or model ambiguity.

515 In this work, we construct the credal set using bootstrap ensembling
516 of the surrogate models (SVR and GBR). Specifically, we train $B = 30$
517 surrogate models per algorithm on independently resampled training sets.
518 The procedure is:

- 519 • Generate B bootstrap resamples from the original training set (sam-
520 pling with replacement).
- 521 • Train one surrogate model on each bootstrap sample.
- 522 • For a new input $\mathbf{x}$, obtain B predictions: $\{f_b(\mathbf{x})\}_{b=1}^B$.

523 This ensemble yields an empirical distribution of predicted values at $\mathbf{x}$.
 524 The credal bounds are then defined as:

$$\underline{f}(\mathbf{x}) = \min_{1 \leq b \leq B} f_b(\mathbf{x}), \quad \bar{f}(\mathbf{x}) = \max_{1 \leq b \leq B} f_b(\mathbf{x}) \quad (11)$$

525 Alternatively, one may define bounds based on central quantiles (e.g.,
 526 5th and 95th percentiles) to reduce sensitivity to outliers. The width of the
 527 credal interval reflects the model’s epistemic uncertainty at input $\mathbf{x}$: narrow
 528 intervals suggest higher confidence, while wider intervals indicate sparse or
 529 extrapolative regions of the input space.

530 By adopting this nonparametric, distribution-agnostic formulation, the
 531 bootstrap ensemble enables robust uncertainty quantification under deep
 532 uncertainty, without relying on strong assumptions about the form of the
 533 predictive distribution.

534 4. Results

535 4.1. Surrogate Models and Performance

536 Machine-learning surrogates were trained to predict capacity tuples from
 537 scour depth and derived features, using 80% of the simulated data points
 538 for each capacity and scour pair. Prior to conducting credal-set uncertainty
 539 quantification, model testing was performed using the remaining 20% of data.
 540 Model performance metrics, as shown in Table 3, confirm that GBR outper-
 541 formed SVR across all capacity components, yielding higher R^2 and lower
 542 RMSE. Further explanations are provided below in relation to their predic-
 543 tive behavior.

Table 3: Surrogate Model Performance Metrics (Test Data)

<i>Model</i>	Δ_y		V_y		M_x		θ_x	
	R ²	RMSE	R ²	RMSE	R ²	RMSE	R ²	RMSE
SVR	0.913	0.023	0.894	24.5	0.876	310.2	0.882	0.0017
GBR	0.948	0.018	0.926	20.3	0.910	265.7	0.907	0.0014

544 4.2. Surrogate Prediction and Uncertainty Quantification

545 Figures 8 and 9 present the surrogate ensemble prediction and uncertainty
546 quantification results. Each plot includes (1) the raw simulated data points
547 and (2) the surrogate model predictions with credal-set bounds across scour
548 depths, including the median of the ensemble output, the lower bounds, and
549 the upper bounds. Key observations are summarized as follows.

- 550 • For the simulated capacity–scour data points ($C \in \{\Delta_y, V_y, M_x, \theta_x\}$
551 vs. S_z), the density of data points decreases as S_z increases, reflect-
552 ing the sampling distribution. At extreme scour depths, data points
553 become sparse. However, overall patterns are highly nonlinear:
 - 554 – For strength-related capacities (V_y, M_x), values show slight in-
555 creases at lower scour depths ($S_z < 7.5$ m). Around $S_z = 7.5$ m,
556 M_x exhibits a jump followed by a decline, while V_y shows mild
557 fluctuation before decreasing. Notably, one-to-many mappings
558 or set-valued nonlinearities emerge near $S_z = 7.5$ m, indicating
559 strong nonlinear behaviors in the FE model.
 - 560 – The displacement responses behave differently. The system-level
561 response Δ_y generally increases with scour depth, indicating global

softening. In contrast, the local rotation capacity θ_x increases initially (for $S_z < 7.5$ m), then declines, also displaying non-smooth and set-valued nonlinear characteristics.

- Model inference is represented by the median of the ensemble predictions. The trend lines effectively capture the nonlinear relationships between each capacity and scour depth for all $C \in \{\Delta_y, V_y, M_x, \theta_x\}$. However, SVR and GBR differ in handling non-smooth and set-valued nonlinearities:
 - As shown in Figure 8, SVR approximates these nonlinearities through smooth, continuous mappings, as expected from the kernel-based formulation which yields infinitely differentiable predictions.
 - As shown in Figure 9, GBR’s predictions are also continuous, but not differentiable. In regions with abrupt nonlinearities, the median predictions exhibit local peaks and discontinuities—an expected consequence of GBR’s piecewise-constant decision-tree architecture.
- The credal-set bounds provide analytical intervals representing epistemic uncertainty in the predicted capacities at each S_z . As seen in Figures 8 and 9, interval widths generally increase with scour depth. Importantly, the upper and lower bounds are asymmetric about the median, as further confirmed in Table 4.

These results indicate that both SVR and GBR effectively approximate the nonlinear relationship between scour depth and structural capacity. GBR

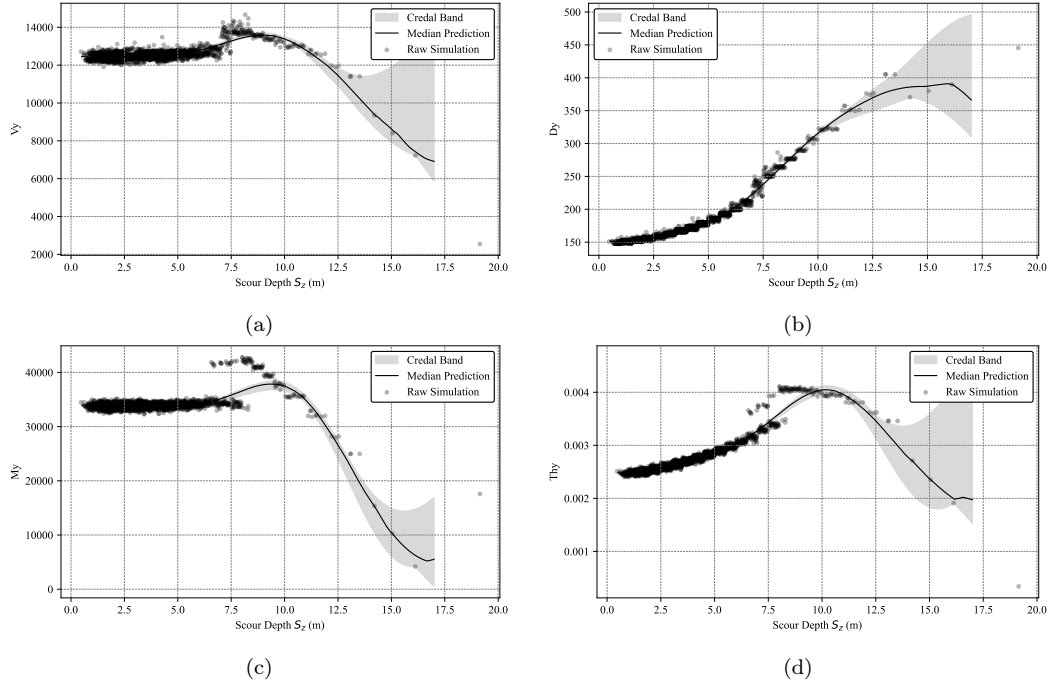


Figure 8: Credal-set surrogate predictions using the SVR models compared against raw simulation data for yield capacities: (a) base shear (V_y), (b) deck displacement (Δ_y), (c) base moment (M_x), and d-rotation (θ_x) as functions of scour depth.

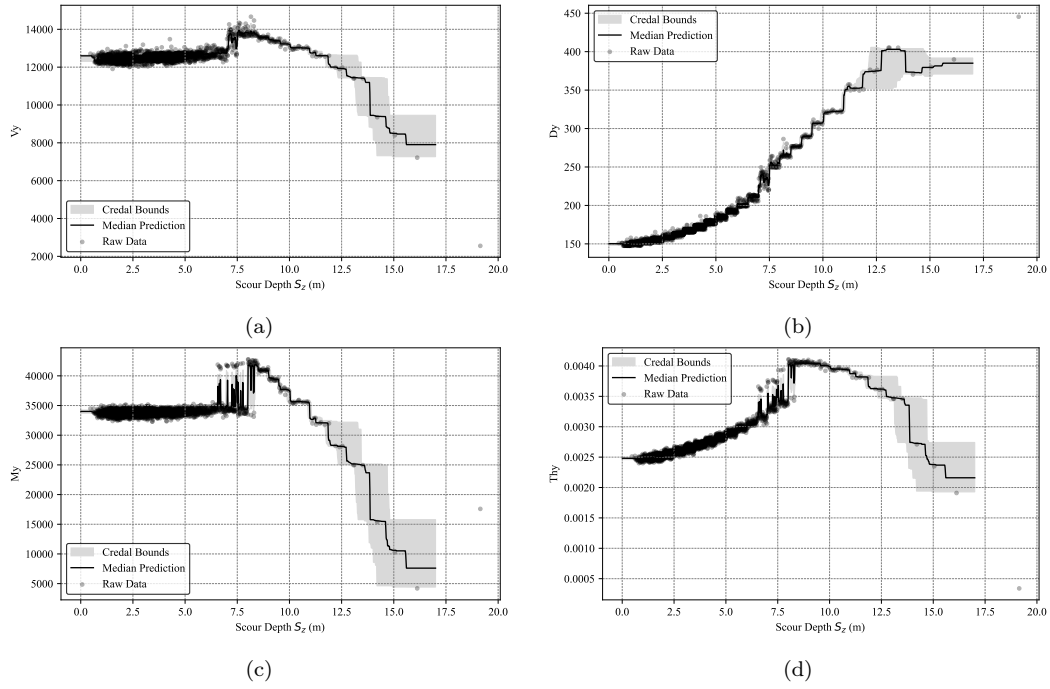


Figure 9: Credal-set surrogate predictions using the GBR models compared against raw simulation data for yield capacities: (a) base shear (V_y), (b) deck displacement (Δ_y), (c) base moment (M_x), and d-rotation (θ_x) as functions of scour depth.

Table 4: Credal-set Bounds for Δ_y and V_y at Key Scour Depths

<i>GBR: Displacement Δ_y (mm)</i>				<i>SVR: Displacement Δ_y (mm)</i>			
<i>Sz (mm)</i>	<i>C_{lb}</i>	<i>C_{med}</i>	<i>C_{ub}</i>	<i>Sz (mm)</i>	<i>C_{lb}</i>	<i>C_{med}</i>	<i>C_{ub}</i>
500	148.78	149.85	150.37	500	164.76	198.47	236.69
1000	148.64	149.44	151.68	1000	148.59	149.69	150.88
5000	174.75	178.46	182.92	5000	181.25	181.51	181.87
10000	305.90	307.51	321.04	10000	309.34	316.61	320.50
15000	371.83	379.42	405.08	15000	272.62	378.04	383.41
<i>GBR: Base Shear V_y (kN)</i>				<i>SVR: Base Shear V_y (kN)</i>			
<i>Sz (mm)</i>	<i>C_{lb}</i>	<i>C_{med}</i>	<i>C_{ub}</i>	<i>Sz (mm)</i>	<i>C_{lb}</i>	<i>C_{med}</i>	<i>C_{ub}</i>
500	12365.31	12444.08	12516.64	500	11257.10	13662.76	15388.41
1000	12328.49	12420.52	12508.12	1000	12332.42	12366.90	12497.20
5000	12311.38	12475.22	12592.03	5000	12582.51	12618.92	12639.30
10000	13016.54	13159.33	13235.98	10000	13050.03	13133.21	13234.92
15000	7346.25	8488.96	11497.43	15000	8313.01	8476.01	11400.96

585 performed better in capturing the capacity transitions when non-smooth and
 586 set-valued relations occur. SVR, using smooth RBF kernels, offered a good
 587 generalization but exhibited limitations in modeling those non-differentiable
 588 relations between the capacity and the scour depth. The credal-set formu-
 589 lation effectively conveys epistemic uncertainty stemming from both model
 590 specification and data sparsity. Prediction intervals widen notably in poorly
 591 sampled regions (i.e., extreme scour), indicating reduced confidence and the
 592 need for conservative estimates in those regimes.

593 *4.3. Feature Dominance Analysis*

594 To interpret surrogate behavior and identify which features most influ-
 595 enced capacity predictions, a feature dominance analysis was conducted for
 596 both SVR and GBR models. The results from both models were largely
 597 consistent (the exception is discussed later). Since the GBR-based interpre-
 598 tation proved more informative due to its superior surrogating performance,
 599 feature dominance results based on the GBR models are described below.

600 Figures 10a to 10d display the GBR-based feature dominance results for
 601 each of the four capacity outputs. These plots illustrate the cumulative con-
 602 tribution of each input feature, where segment count represents the number
 603 of times a feature was selected as a decision-splitting node across all trees
 604 in the ensemble. A higher count indicates a greater influence on predictive
 605 decisions. Figure 11a to 11d visualizes the most dominant feature as the
 606 scour depth varies based on the GBR model as well.

607 The analysis reveals that the logarithmic transformation of scour depth
 608 ($\log Scour$) or $\log(S_z)$, was the most dominant feature across all capacity
 609 components. Other engineered features such as inverse scour depth ($1/S_z$),

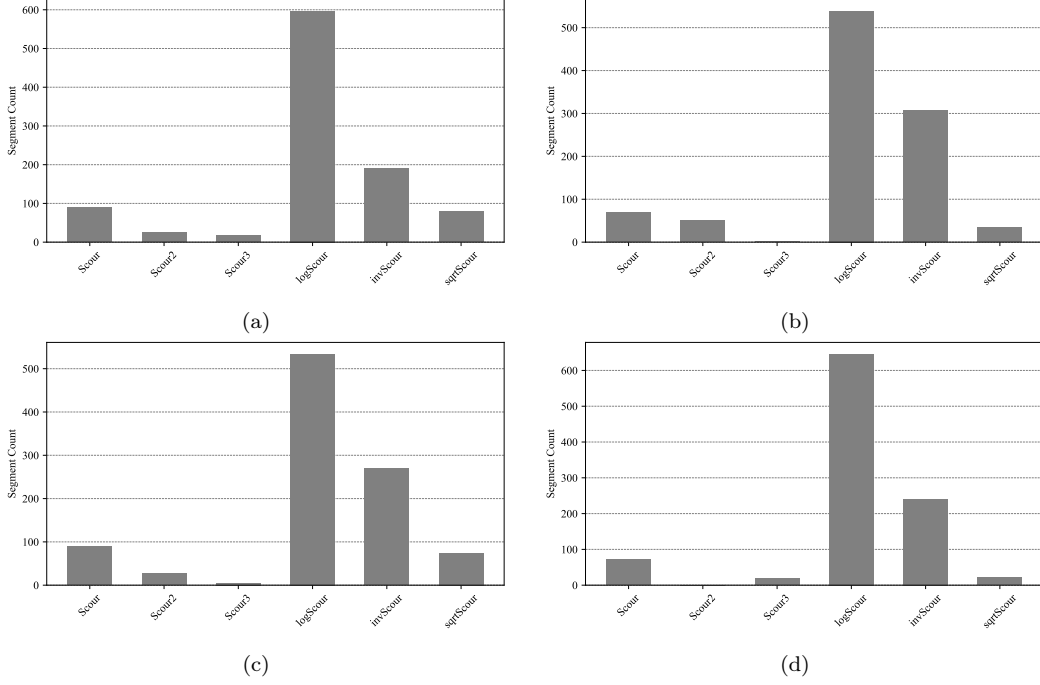


Figure 10: Bar charts of feature dominance counts for each capacity measure, indicating the frequency with which each scour-derived input dominates predictions in the GBR surrogate models: (a) V_y , (b) Δ_y , (c) M_x , and (d) θ_x . Note that the input features of *Scour*, *Scour2*, *Scour3*, *logScour*, *invScour*, and *sqrtScour*, representing S_z : S_z^2 , S_z^3 , $\log(S_z)$, $1/S_z$, and $\sqrt{S_z}$, respectively.

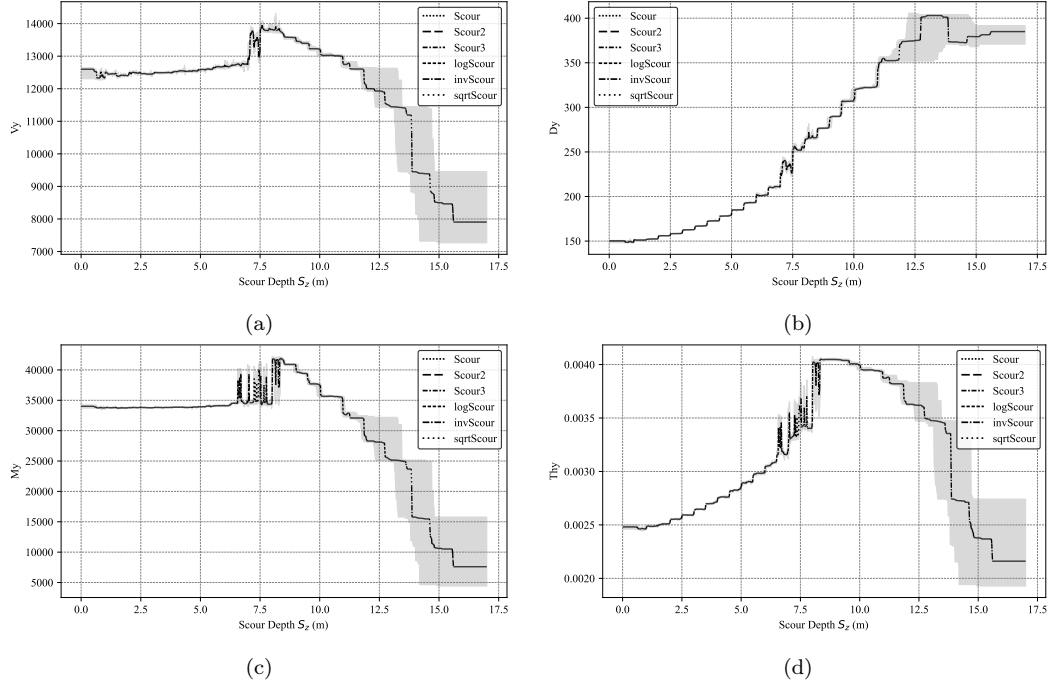


Figure 11: Scour-depth varying visualization of most dominant features for target capacity based on the GBR surrogate models: (a) V_y , (b) Δ_y , (c) M_x , and (d) θ_x . Note that the input features of *Scour*, *Scour2*, *Scour3*, *logScour*, *InvScour*, and *sqrtScour*, representing S_z : S_z^2 , S_z^3 , $\log(S_z)$, $1/S_z$, and $\sqrt{S_z}$, respectively.

square root of scour depth ($\sqrt{S_z}$), and quadratic terms (S_z^2) gained prominence, particularly at larger scour depths, where capacity responses exhibit asymptotic or nonlinear decline. These findings support the inclusion of a diverse set of engineered features to enhance surrogate model expressiveness across different scour regimes.

5. Discussion

The proposed surrogate modeling framework, integrated with credal-set uncertainty quantification, provides a computationally efficient and conservative approach for evaluating bridge capacity degradation under deep, climate-driven uncertainties. We highlight several limitations, practical implications, and directions for future work, as summarized below.

- *Limitation 1: Scope of Uncertainty Representation.* In this work, scour is treated as the dominant causal source of uncertainty when assessing bridge capacity degradation under climate scenarios. Material uncertainties (i.e., concrete compressive strength and steel yield strength) were incorporated in the FE simulations as random variables but were treated as background fluctuations rather than causal drivers of scour effects. We recognize that, in a more comprehensive climate-change context, variables such as ambient temperature, moisture, and corrosion may lead to progressive material degradation and thus partially and causally affect bridge capacity. In such cases, a multi-input surrogate modeling framework that integrates both environmental and material factors would become essential for accurate and generalizable prediction.

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partitioning.

- *Implication 2: Extensibility to Capacity–Demand Analysis.* The current framework focuses on modeling intrinsic structural capacities under scour. However, understanding collapse vulnerability necessitates integrating extrinsic response demands, such as those from seismic loading or barge impacts. These demands are inherently stochastic and scenario-dependent. Future work should consider extending the surrogate approach to model both capacities and demands, enabling reliability-based assessment using credal-set formulations that avoid restrictive distributional assumptions.
- *Practical Value and Future Extension.* The use of surrogate learning and credal-set intervals to frame capacity prediction—namely, in terms of $(C_{lb}, C_{med}, C_{ub})$ rather than a single-valued prediction—offers an informed and interpretable framework for risk-informed decision-making. This is particularly relevant in resilience planning, where high-stakes infrastructure decisions must be made under poorly understood or evolving environmental conditions. Moving forward, integrating credal-based predictions into decision models that account for socioeconomic and behavioral factors (e.g., through multi-criteria or game-theoretic frameworks) represents an important avenue for interdisciplinary extension.

680 6. Conclusion

681 This study developed and demonstrated a surrogate-based framework for
682 evaluating the yield-level capacity degradation of scour-critical bridges un-
683 der deeply uncertain climate-driven scour scenarios. By integrating nonlinear
684 pushover simulations with machine learning surrogates and credal-set uncer-
685 tainty quantification, the framework enables both computational efficiency
686 and distribution-agnostic capacity estimation.

687 Specifically, a suite of nonlinear finite-element analyses was conducted
688 across 3,000 sampled scour conditions, generating a large-scale dataset of
689 capacity responses. These responses were organized as a multi-dimensional
690 capacity tuple (deck displacement, base shear, column moment, and rota-
691 tion) at yield to characterize both superstructure and substructure perfor-
692 mance. Two classes of surrogate models, Support Vector Regression (SVR)
693 and Gradient Boosting Regression (GBR) were trained using both raw and
694 engineered features of scour depth to emulate the simulation outputs. Fea-
695 ture engineering, especially logarithmic and inverse transformations of scour
696 depth, was shown to be critical for capturing nonlinear capacity trends.

697 To address deep epistemic uncertainty associated with future scour sever-
698 ity, a credal-set formulation was employed using bootstrap ensembles. For
699 each surrogate model, bootstrapped resamples were trained and evaluated,
700 yielding lower and upper bounds on capacity predictions for any input scour
701 depth. These credal bounds formed objective and distribution-agnostic in-
702 tervals suitable for decision-making under high uncertainty. Comparative
703 performance evaluation showed that GBR consistently outperformed SVR in
704 prediction accuracy across all capacity metrics, while SVR provided smoother

705 uncertainty envelopes in the scour-depth regime.

706 The results demonstrate that the proposed framework not only reduces
707 computational cost but also explicitly accommodates structural and environ-
708 mental uncertainty in a way that is practical and scalable. The methodology
709 advances current practice by framing capacity prediction as an interval-based
710 decision support tool and offers a path forward for risk-informed infrastruc-
711 ture assessment under progressive climate impacts.

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